

HARBOR BEACH, MI, USA

WMO: 997260

Lat: 43.850N

Lon: 82.640W

Elev: 588

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 1.4	(c) 5.5	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 47.3	(k) 19.4	(l) 37.5	(m) 22.1	(n) 11.8	(o) 250	(p) 0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.8	(c) 84.2	(d) N/A	(e) 80.7	(f) N/A	(g) 77.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 13.2	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 34.7	(b) 28.4	(c) 24.3	(e) -3.5	(f) 89.3	(g) 7.4	(h) 3.3	(i) -8.8	(j) 91.7	(k) -13.1	(l) 93.6	(m) -17.3	(n) 95.4	(o) -22.7	(p) 97.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.3	22.5	23.1	31.3	41.9	52.5	62.0	69.1	68.3	62.1	51.4	39.4	30.6		
	DBStd	18.01	9.92	9.44	8.95	6.92	7.69	6.72	5.88	4.75	6.25	7.24	8.12	7.73		
	HDD50	3516	852	754	584	261	61	2	0	0	1	70	329	602		
	HDD65	7194	1317	1174	1046	694	394	135	26	18	127	427	769	1067		
	CDD50	2168	0	0	3	16	140	363	592	568	364	112	10	0		
	CDD65	371	0	0	0	0	7	46	153	121	40	4	0	0		
	CDH74	1602	0	0	0	5	52	253	719	407	144	22	0	0		
	CDH80	369	0	0	0	0	7	61	182	82	31	6	0	0		
(14) Wind	WSAvg	11.5	12.9	12.5	12.9	13.4	11.0	10.5	9.3	9.7	9.9	11.6	11.5	13.3		
(15) Precipitation	PrecAvg	31.9	2.0	1.9	1.8	3.0	3.4	3.1	3.1	2.9	3.2	2.7	2.4	2.2		
	PrecMax	40.2	3.7	5.5	4.5	5.6	6.6	6.2	5.4	4.8	7.4	5.5	5.6	4.1		
	PrecMin	24.2	0.6	0.5	0.7	0.9	1.7	1.0	0.9	0.6	0.4	1.6	0.7	0.9		
	PrecStd	3.7	0.9	1.1	1.0	1.3	1.5	1.5	1.1	1.0	1.5	1.1	1.2	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.4	50.0	62.1	71.4	80.9	86.5	89.6	86.9	83.8	76.2	62.4	53.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	44.4	42.2	54.3	62.3	74.4	80.6	84.7	81.9	78.2	68.9	57.4	47.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	39.1	37.6	46.5	55.8	69.8	76.5	81.4	78.2	73.8	64.6	54.0	43.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	35.6	34.4	41.8	50.9	65.3	72.1	77.8	75.4	71.0	61.4	50.8	40.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	10.7	10.4	10.8	12.8	11.4	10.8	10.2	10.7	10.3	9.1	8.5		
		MCDBR	14.5	13.9	18.4	21.3	21.6	18.8	16.6	14.5	15.1	15.0	14.4	12.6		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.306	0.307	0.316	0.362	0.386	0.414	0.411	0.402	0.373	0.356	0.318	0.287		
	taud		2.392	2.365	2.413	2.345	2.351	2.297	2.340	2.352	2.423	2.438	2.509	2.508		
	Ebn at Noon		259	281	293	286	281	272	272	270	271	260	255	259		
	Edh at Noon		24	30	32	37	38	41	39	37	32	28	22	20		
(44) All-Sky Solar Radiation	RadAvg		378	641	1088	1444	1708	1919	1909	1644	1298	797	458	295		
	RadStd		43	69	110	126	148	111	88	77	121	78	65	34		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page